

End-to-End Data Analytics Case Study

Project Objective

Analyze the Sample Superstore dataset from a business perspective by defining the business problem, calculating KPIs, performing sales analysis, generating insights, and providing actionable recommendations.

Business Problem

The management wants to understand overall business performance, identify the best-performing regions, categories, customer segments, and profitable products to support strategic decision-making.

Analysis Performed

- Business Problem Definition
- KPI Calculation (Sales, Profit, Orders, Quantity, Discount, Profit Margin)
- Sales by Region Analysis
- Sales by Category Analysis
- Customer Segment Analysis
- Profitability Analysis (Sub-Categories)
- Executive Business Insights

Key Findings

- Top Performing Region: **West**
- Top Performing Category: **Technology**
- Top Customer Segment: **Consumer**
- Most Profitable Sub-Category: **Copiers**

Business Recommendations

1. Increase investment in the West region.
2. Focus on Technology products for business growth.
3. Improve performance in lower-performing regions.
4. Strengthen engagement with Corporate and Home Office customers.
5. Continue promoting high-profit sub-categories such as Copiers.
6. Monitor discounts carefully to protect profit margins.

Conclusion

This case study demonstrates a complete end-to-end data analytics workflow, transforming raw business data into meaningful insights that support data-driven decision-making.